Department of Computer Science and Engineering (CSE)
BRAC University

Practice Problem Set 3.3

CSE251 - Electronic Devices and Circuits

DIODE LOGIC CIRCUITS

OR Gate, AND Gate, and Cascaded Logic Circuits

Course Description, COs, and Policies

Midterm and Final Questions

For any error or issues, contact purbayan.das@bracu.ac.bd

Problem 1

- Implement the following logic functions using diodes. Here, A, B, C, and D are Boolean inputs. Here "+" and "." denotes logical OR and AND, respectively.

  I. $A.C + B + D$ [Fall 2024]

  II. $A.C + B + D$ [Fall 2024]

  III. $(A + B).(C + D)$ [Fall 2023]

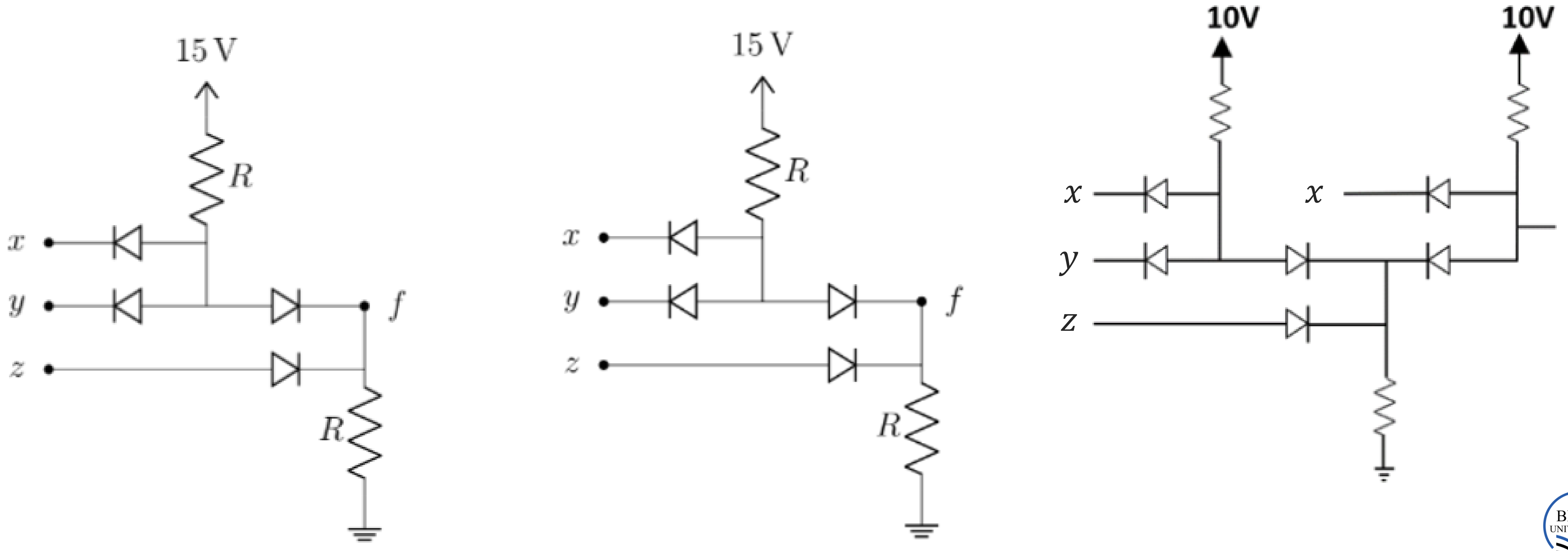
  IV. $(A + B).C + D.E$ [Fall 2025]

  V. $A.B.C + D$ [Spring 2025]

  VI. $(A + B + C).D$ [Spring 2025]

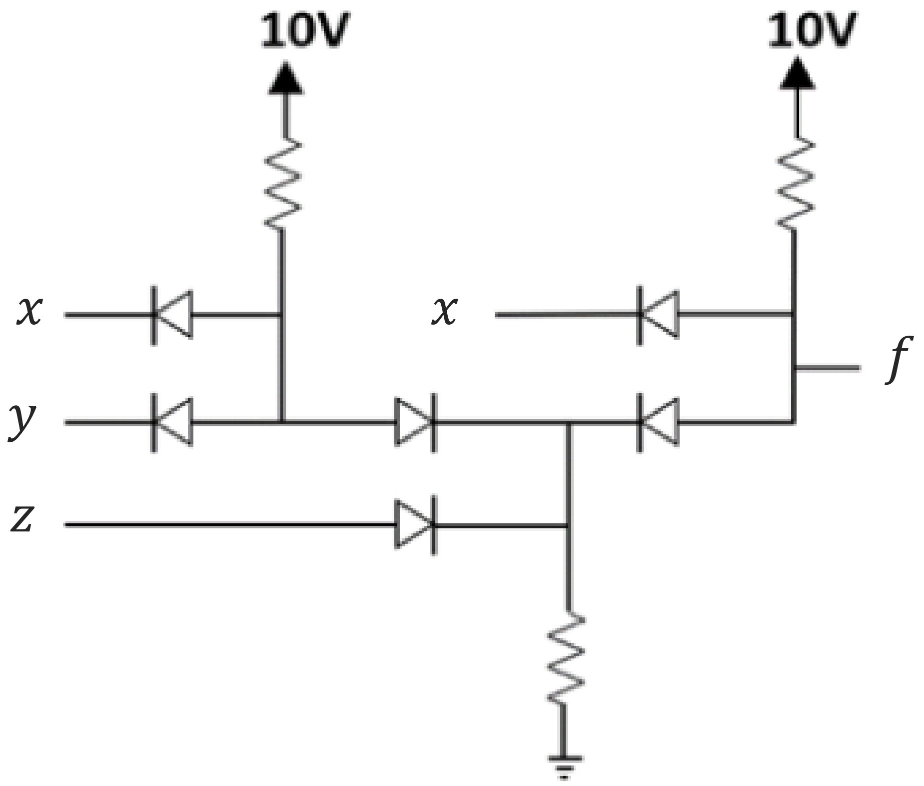
Problem 2

- For each of the circuits shown below, assuming x, y, and z are Boolean inputs, express f in terms of x, y, and z.



Problem 3

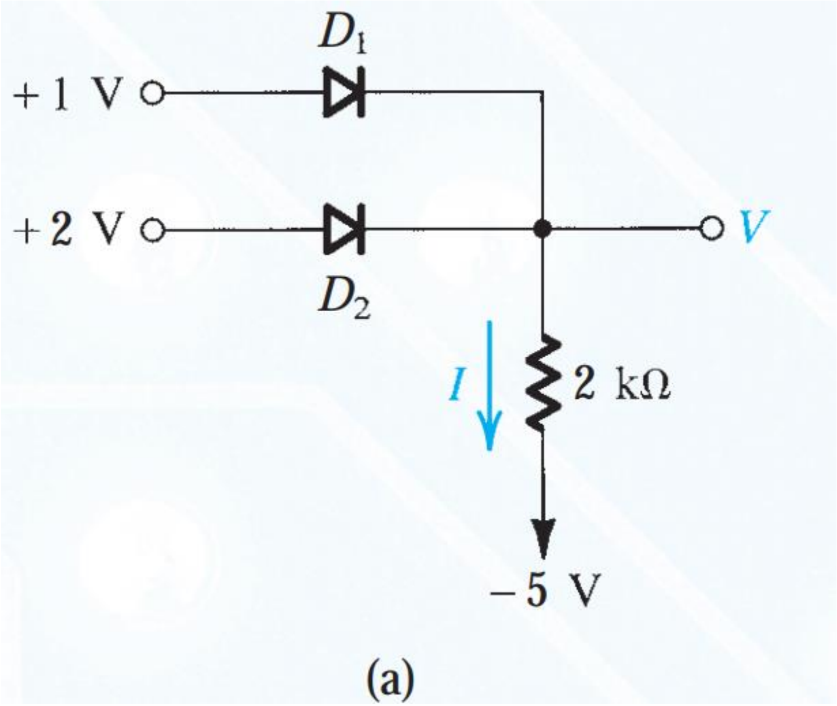
- If x, y, and z are Boolean inputs and f is the Boolean output, complete the following table. Assume that the diodes are ideal.

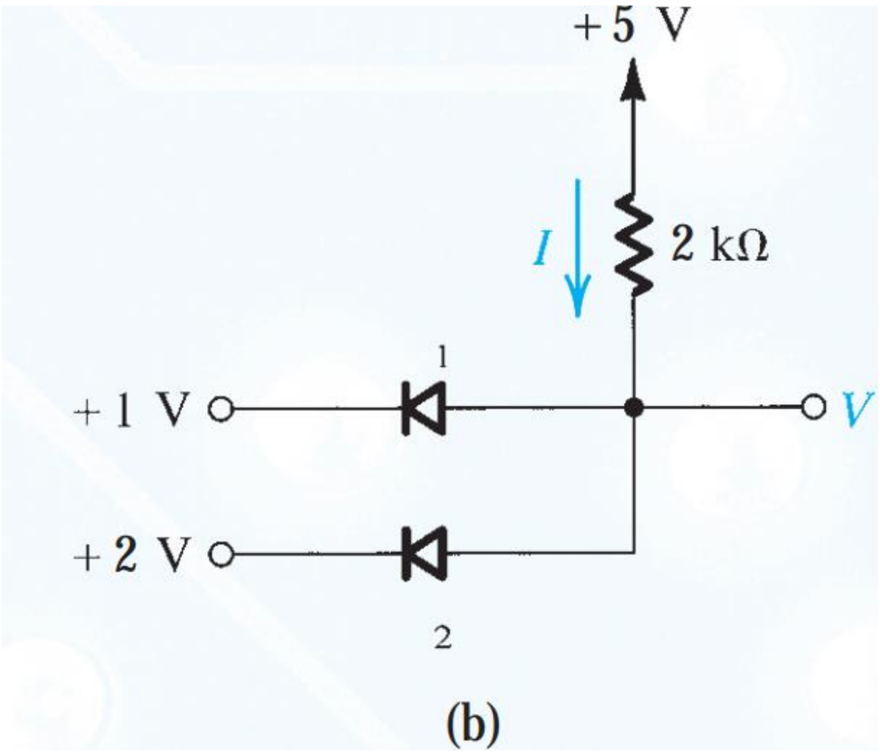


x	y	z	f
$4\ V$	$3\ V$	$4\ V$	
$5\ V$	$3\ V$	$4\ V$	
$13\ V$	$3\ V$	$4\ V$	

Problem 4

- For the logic circuits shown below, use ideal diode model, find the values of the voltages and currents indicated.

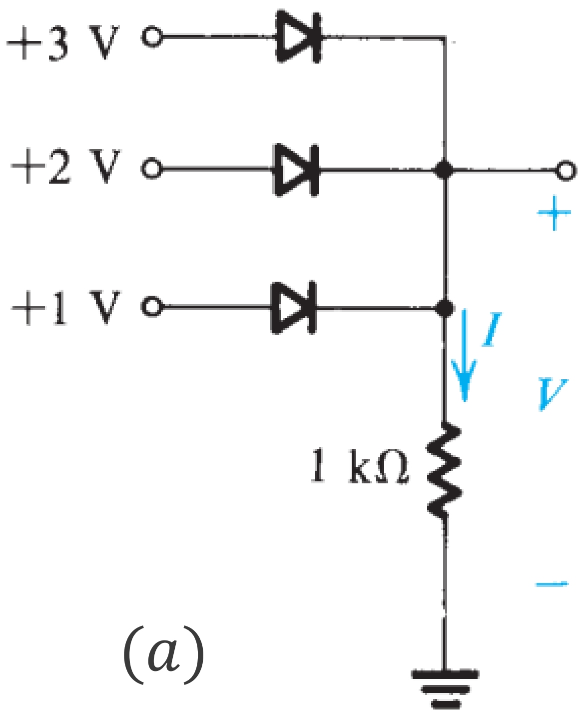




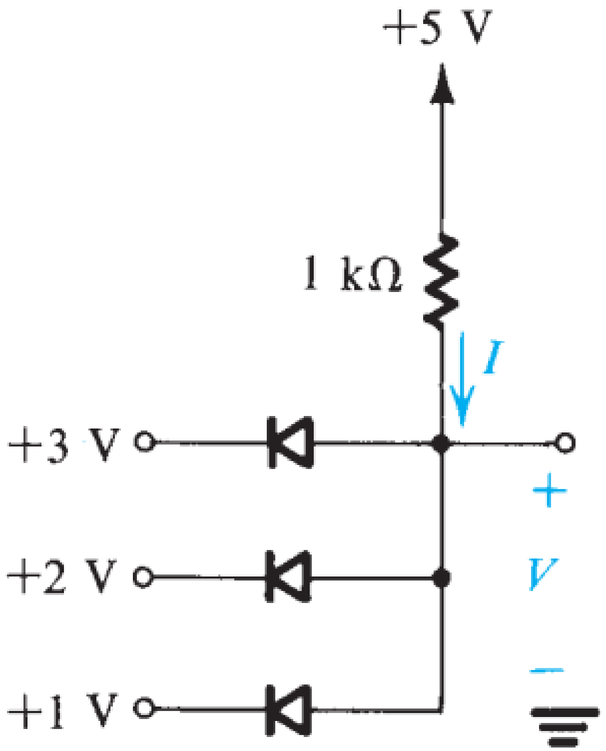
Ans: $(a)\ V = 2\ V,\ \ I = 2.5\ mA;\ (b)\ V = 1\ V,\ \ I = 2\ mA$

Problem 5

- For the logic circuits shown below, use ideal diode model, find the values of the voltages and currents indicated.



(a)

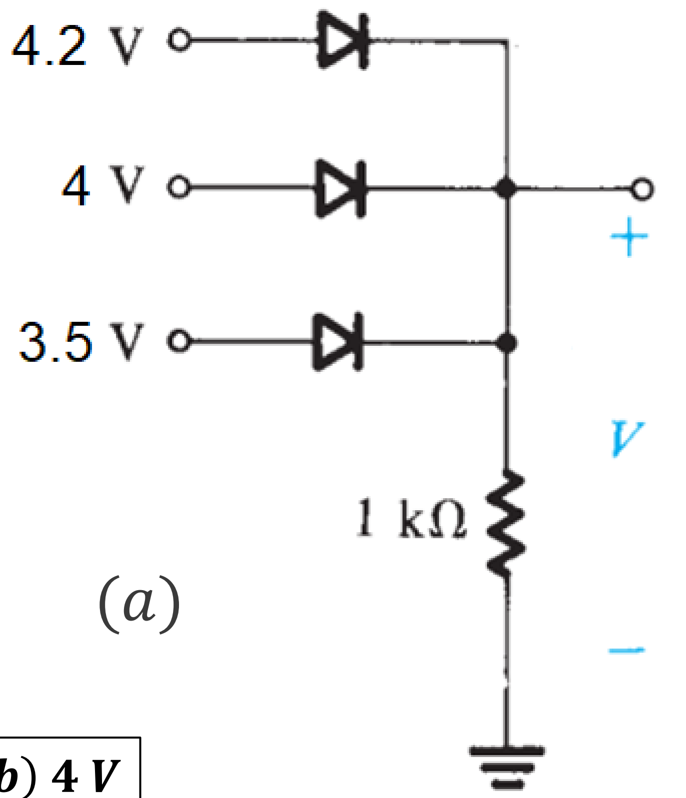


(b)

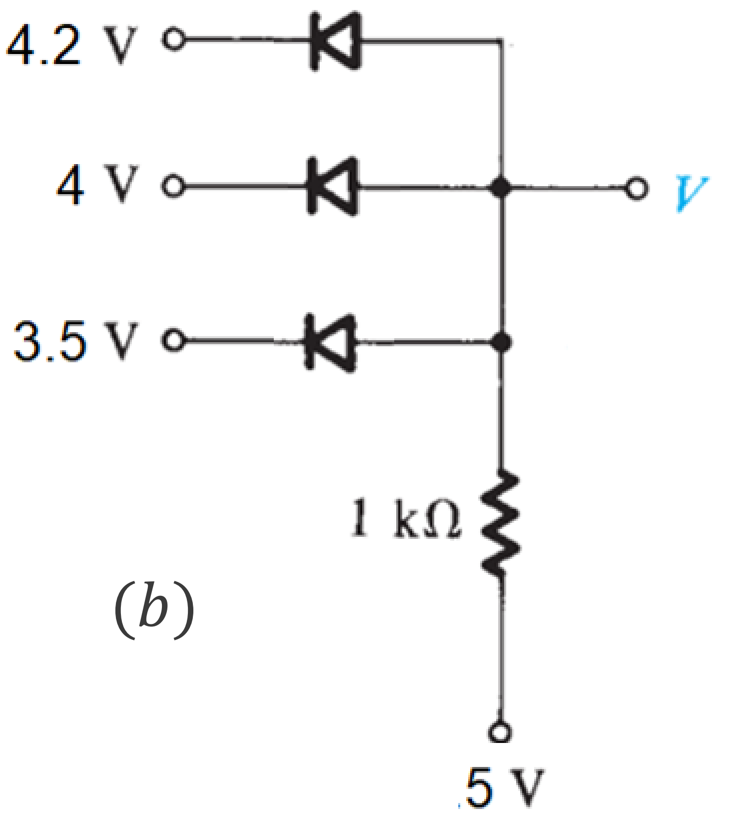
Ans: $(\boldsymbol{a})\ \boldsymbol{V = 3\ V,\ \ I = 3\ mA;}\ (\boldsymbol{b})\ \boldsymbol{V = 1\ V,\ \ I = 4\ mA}$

Problem 6

- For each of the circuits shown below, use CVD model, find the values of the voltages and currents indicated. Take $V_{D_{o_1}} = 1\,V$, $V_{D_{o_2}} = 0.7\,V$, and $V_{D_{o_3}} = 0.5\,V$.



(a)

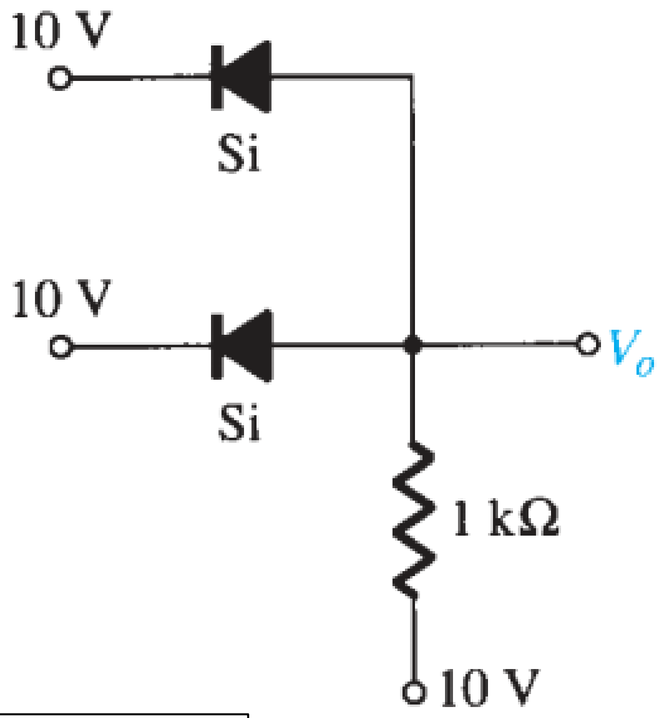


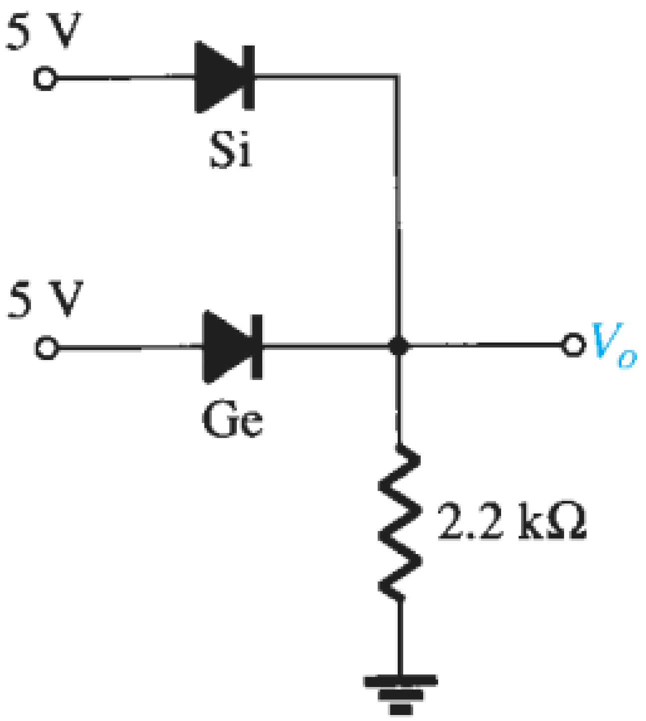
(b)

Ans: (a) $\boldsymbol{V = 3.3\ V}$; $\boldsymbol{(b)\ 4\ V}$

Problem 7

- For the logic circuits shown below, determine. Use CVD+R model with $V_{D_o,\,Si} = 0.7\ V$ and $V_{D_o,\,Ge} = 0.3\ V$ and $r_o = 10\ \Omega$ for both.

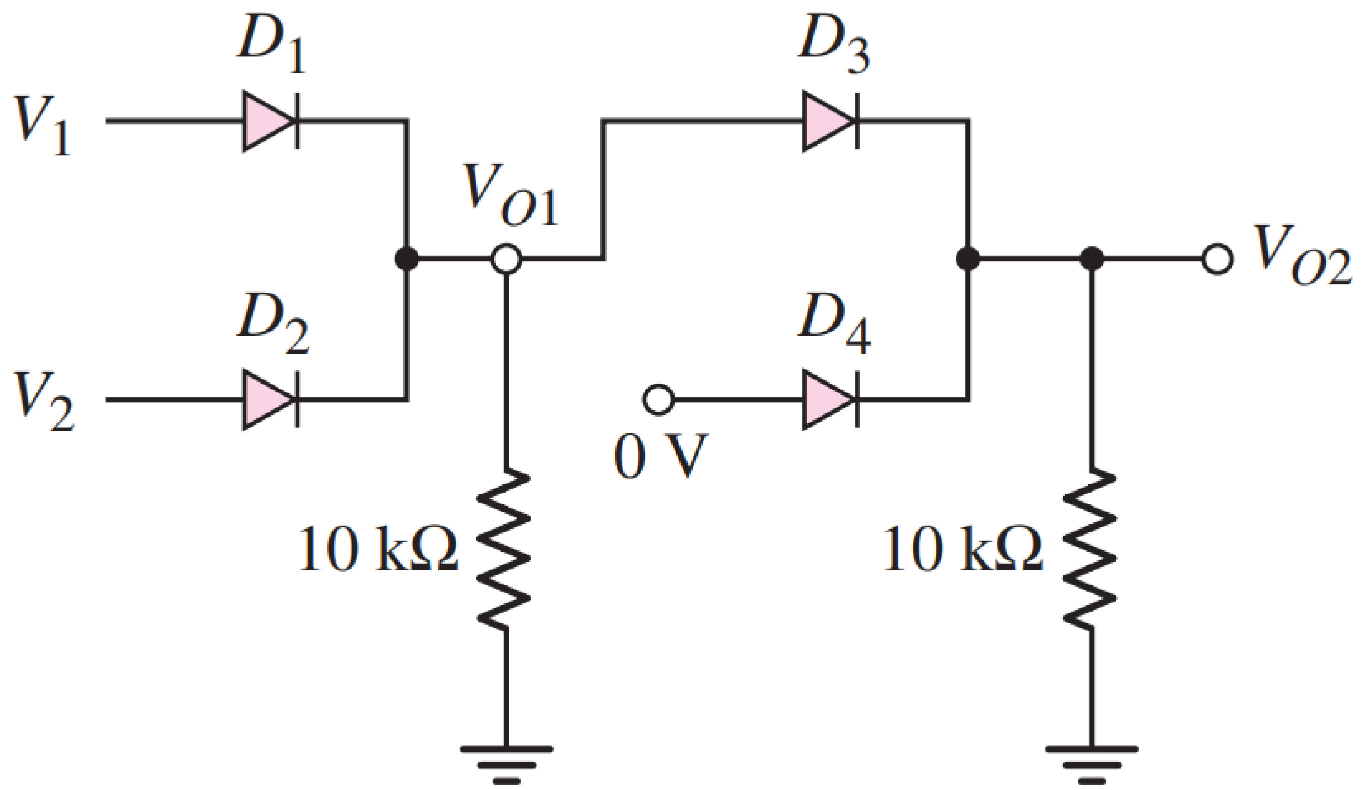




<u>Ans</u>: $(\boldsymbol{a})\ \boldsymbol{V_o = 10.7\ V};\ (\boldsymbol{b})\ \boldsymbol{V_o = 4.7\ V}$

Problem 8

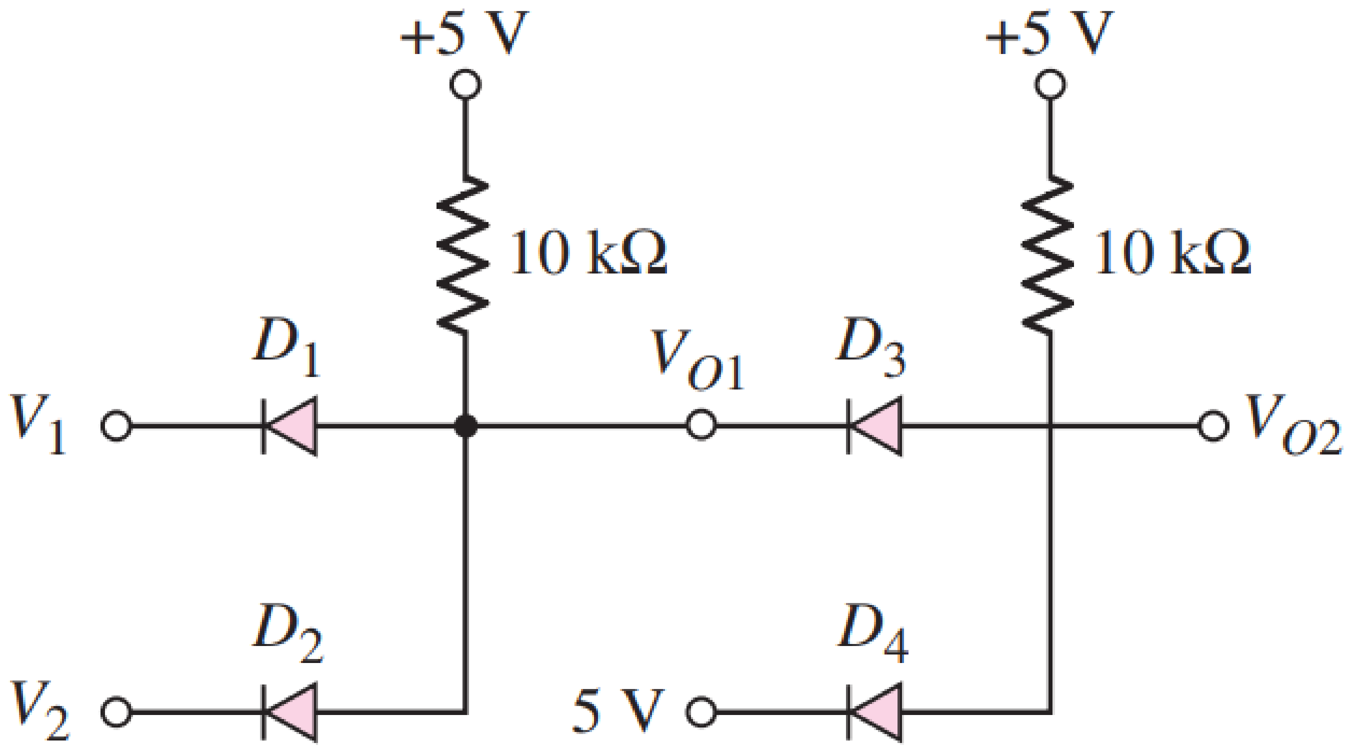
- From the logic network shown, express V_{O2} as a Boolean expression of V_1 and V_2. Determine V_{O2} if $V_1 = 3\ V$ and $V_2 = 1\ V$. Use CVD model with $V_{D_o} = 0.7\ V$.



Ans: $\boldsymbol{V_{O2} = 2\ V}$

Problem 9

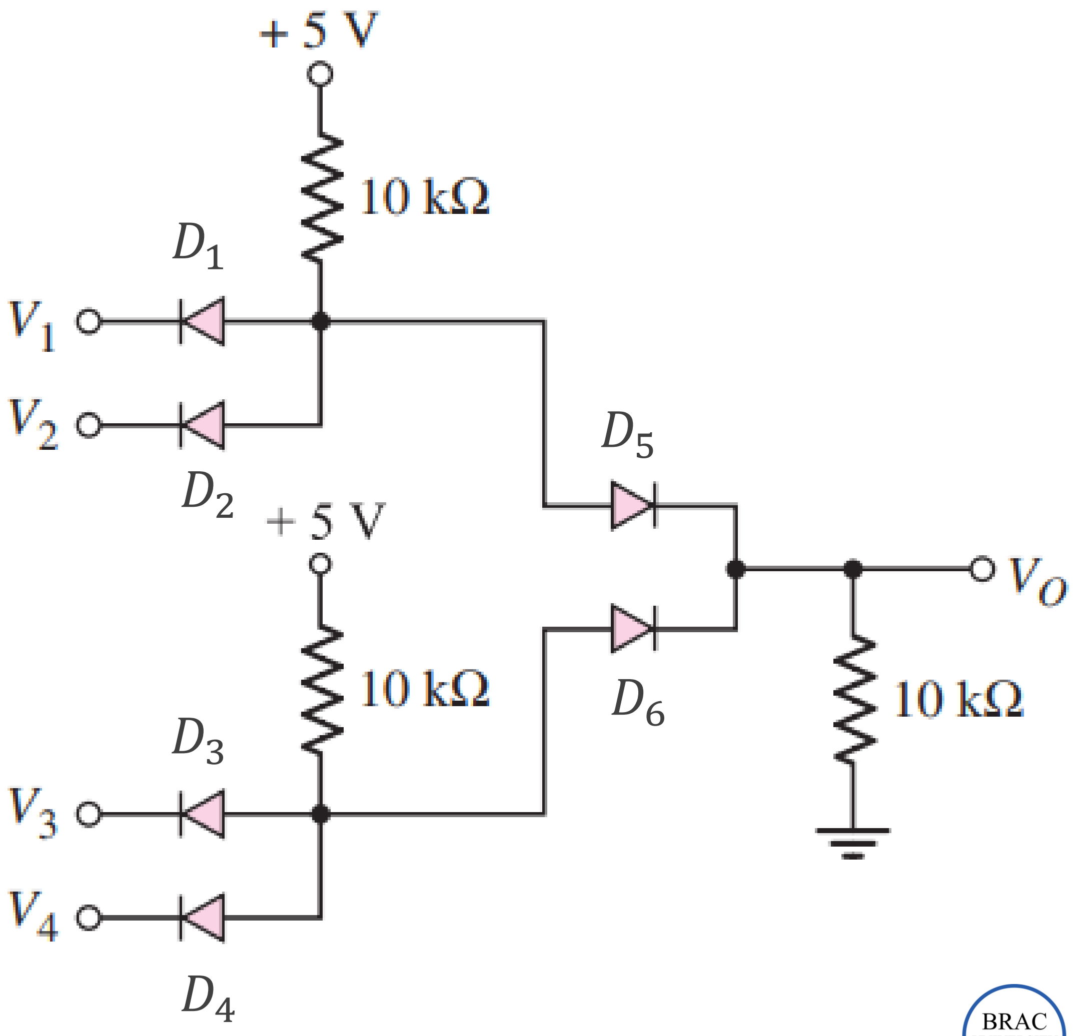
- From the logic network shown, express V_{O2} as a Boolean expression of V_1 and V_2. Determine V_{O2} if $V_1 = 3\ V$ and $V_2 = 1\ V$. Use CVD model with $V_{D_o} = 0.7\ V$.



Ans: $\boldsymbol{V_{O2} = 2.4\ V}$

Problem 10

- From the logic network shown, determine V_O if $V_1 = 2\,V$, $V_2 = 2.2\,V$, $V_3 = 2.4\,V$, and $V_4 = 2.5\,V$. Use CVD model with $V_{Do_1} = 0.3\,V$, $V_{Do_2} = 0.5\,V$, $V_{Do_3} = 0.7\,V$, $V_{Do_4} = 0.9\,V$, $V_{Do_5} = 1\,V$, and $V_{Do_6} = 1.5\,V$.



Ans: $\boldsymbol{V_O = 1.6\,V}$

Up Next

Relevant Questions from Past Semesters

Mid – Fall 2025

In a diode logic circuit, a logic HIGH is approximately $1.8\ V$, while a logic LOW is approximately $0\ V$. The circuit is designed to implement the Boolean function

$$f = w\,.\,(x + y).\,z,$$

where, w, x, y, and z are the Boolean inputs and f is the Boolean output.

a. Draw the diode logic circuit that realizes the Boolean function given above.

b. For this logic circuit, determine the output voltage corresponding to the input values $w = 0.3\ V$, $x = 0.7\ V$, $y = 1.7\ V$, and $z = 1.9\ V$. Assume all diodes are identical with negligible resistance and have a cut-in voltage of $0.7\ V$.

Ans: $(\boldsymbol{b})\ \boldsymbol{f = 1\ V}$

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Mid – Summer 2025

If x, y, and z are Boolean inputs and f is the output,

a. Design a circuit using diode logic to implement the following Boolean function:

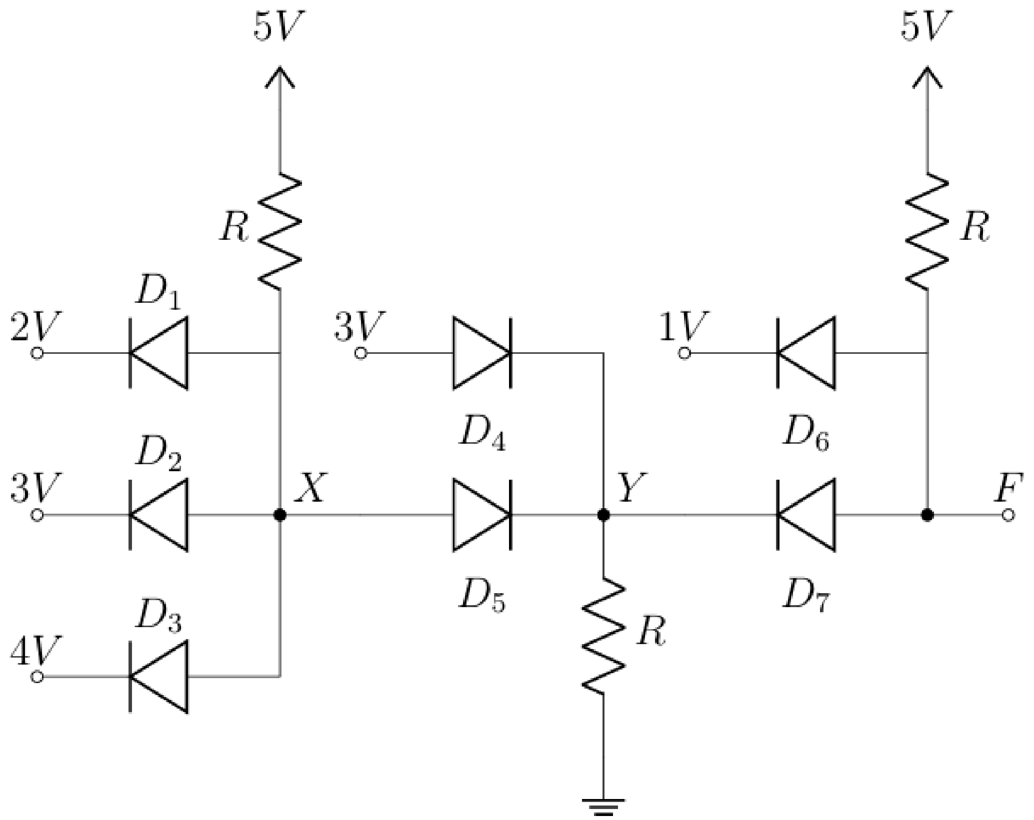
$$f = x.y + z$$

b. Given the input voltages $x = 3.3\ V$, $y = 3.0\ V$, and $z = 2.7\ V$, determine the output voltage of the logic circuit designed in (a). Assume a supply voltage of $5\ V$, where applicable. Also, assume that the diodes have a cut-in voltage of $V_{D_0} = 0.7\ V$ and negligible resistance compared to other components.

Ans: $(\boldsymbol{b})\ \boldsymbol{f} = \mathbf{3\ V}$

Mid – Spring 2025

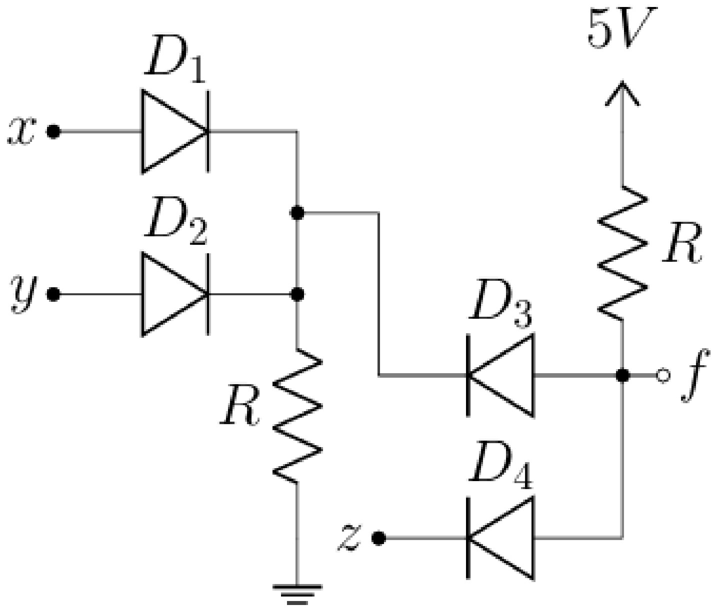
- Assuming that all the diodes are ideal on the following circuit, determine V_F.



Ans: $\boldsymbol{V_F = 1\ V}$

Mid – Fall 2024

- If x, y, and z are the Boolean inputs and f is the Boolean output, for the logic circuit shown below, write an expression for f.



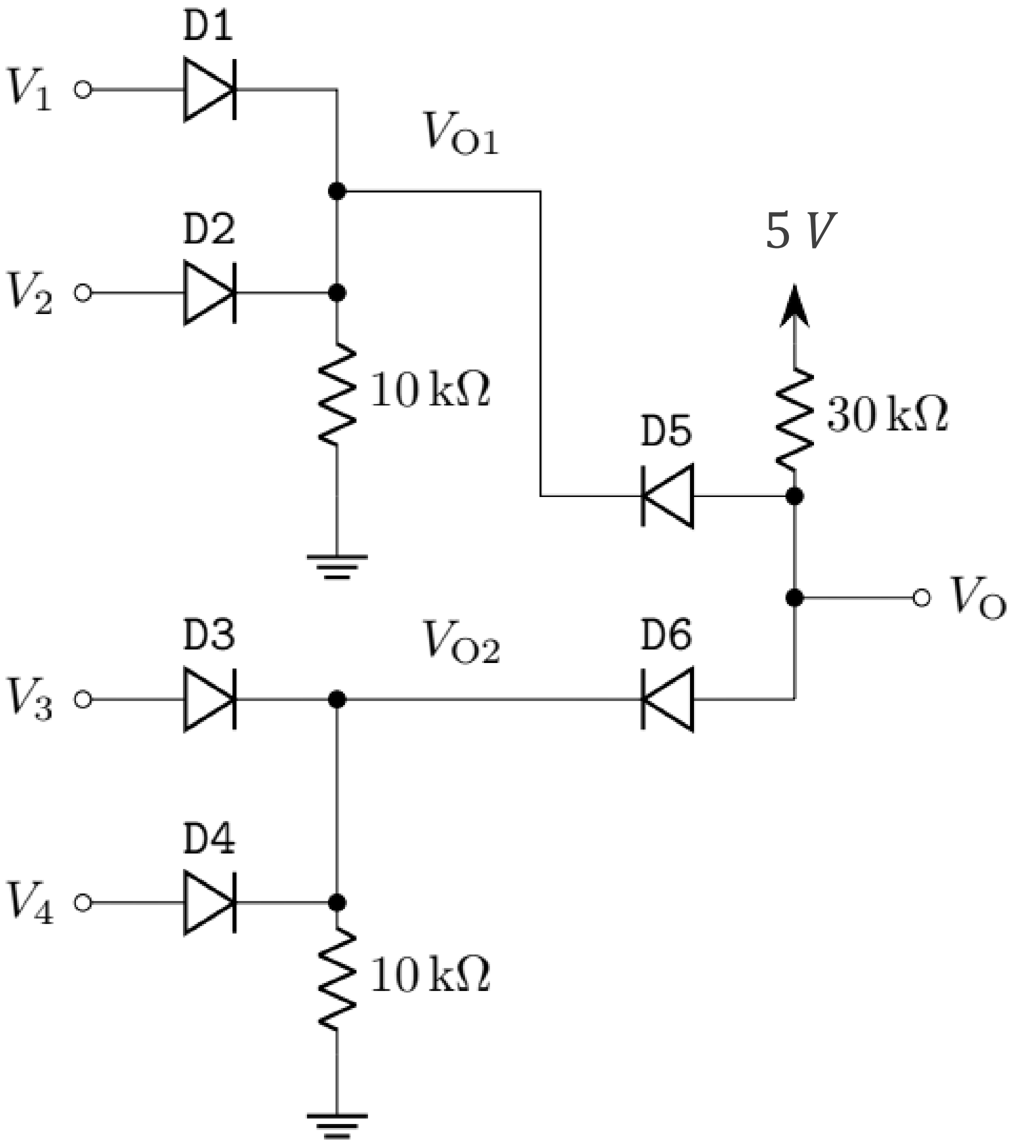
Ans: $f = (x + y) \cdot z$

Mid – Summer 2024

a. From the logic network shown, determine V_{O1}, V_{O2}, and V_O if $V_1 = 2\,V$, $V_2 = 2.2\,V$, $V_3 = 2.4\,V$, and $V_4 = 2.5\,V$. Use CVD model with $V_{D_{o1}} = 0.3\,V$, $V_{D_{o2}} = 0.7\,V$, $V_{D_{o3}} = 0.5\,V$, $V_{D_{o4}} = 0.9\,V$, and $V_{D_{o5}} = V_{D_{o6}} = 1\,V$.

b. Now if V_3 and V_4 are changed to $-2\,V$ and $-3\,V$ while keeping all other parameters unchanged, identify the states of the diodes D_3 and D_4. Determine V_O.

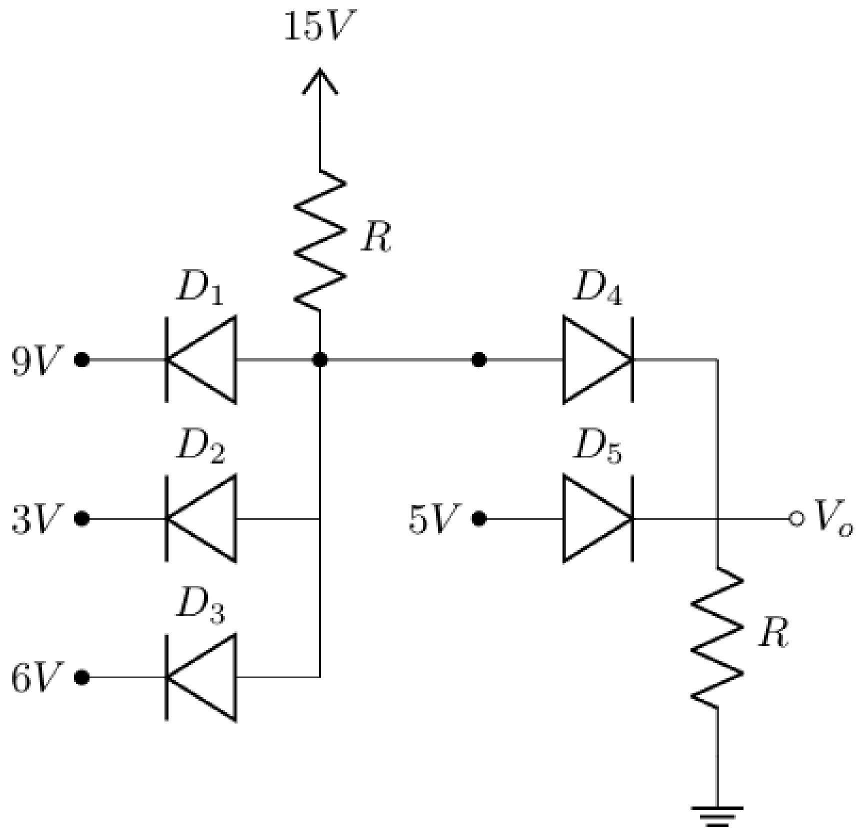
[Hint: V_{O2} cannot be calculated as in (a) for negative V_3 and V_4. You need to apply method of assumed states to check the states of $D5$ and $D6$.]



Ans: $(a)\ \boldsymbol{V_{O1} = 1.7\,V,\ V_{O2} = 1.9\,V, V_O = 2.7\,V}$; $(b)\ \boldsymbol{V_{O1} = 1.7\,V,\ D_5 \rightarrow OFF, D_6 \rightarrow ON, I_{D_6} = 0.1\,mA, V_{D_5} = 0.3\,V, V_O = 2\,V}$

Mid – Summer 2023

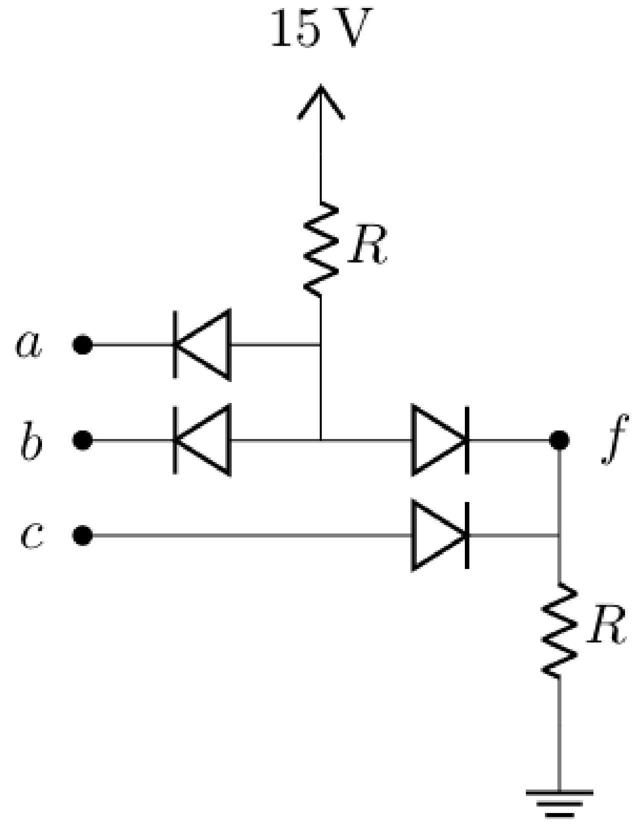
- Assuming all the diodes in the following logic circuit are ideal, determine V_o.



Ans: $V_o = 5\ V$

Mid – Spring 2023

- If a, b, and c are the Boolean inputs and f is the Boolean output, for the logic circuit shown below, write an expression for f in terms of the inputs.

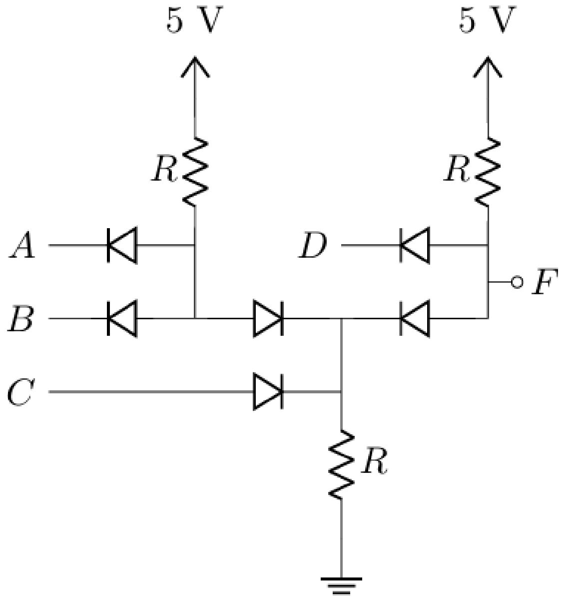


Ans: $f = a.b + c$

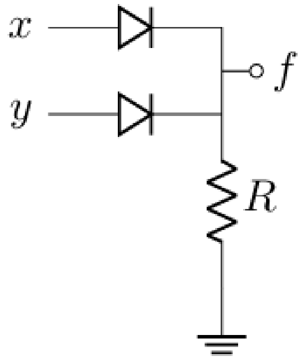
Mid – Fall 2022

a. If A, B, C, and D are the Boolean inputs and F is the Boolean output, for the Circuit-1, write an expression for F.

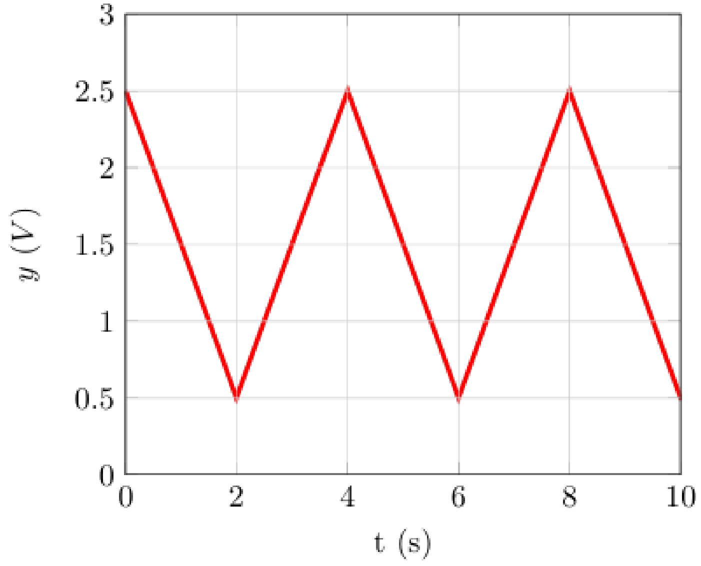
b. Draw a voltage vs. time graph for the voltage at f assuming x and y are voltage signals, where $x = 1.5\ V$ and y has a waveform as shown in the Graph-1.



Circuit-1



Circuit-2



Graph-1

Ans: (a) $F = (A.B + C).D$

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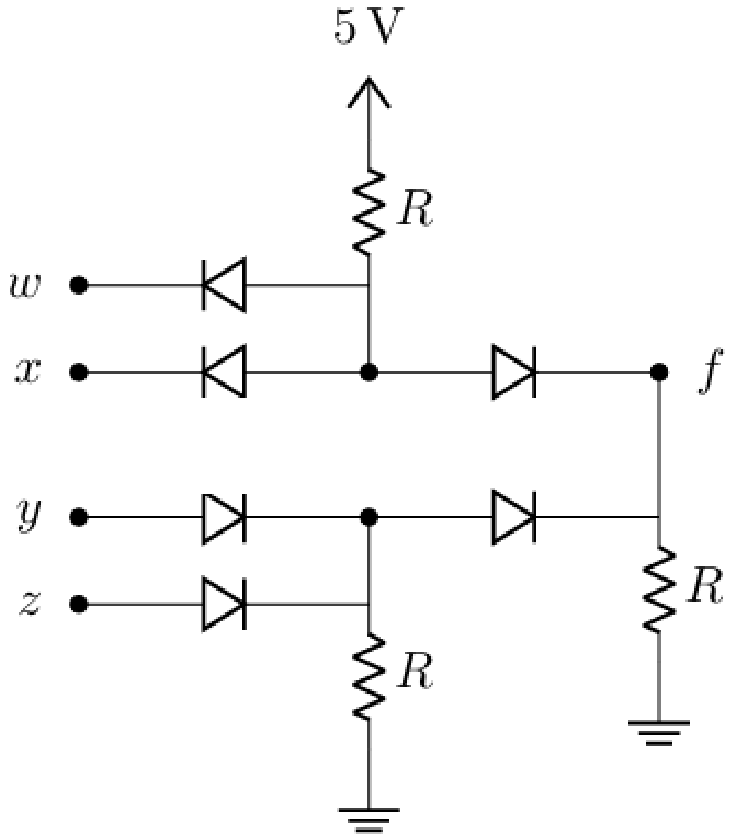
Mid – Summer 2022

a. If w, x, y, and z are the Boolean inputs and f is the Boolean output, for the logic circuit shown below, write an expression for f in terms of the inputs.

b. Jawad has created a new ride sharing app- $Juber$. When you request a Juber ride, Juber's algorithm generates 4 signals to determine whether it will be forwarded to a rider. (1) Signal F determines if the rider is free. (2) Signal R determines if the rider is within close proximity. (3) Signal G determines if the rider has good rating. (4) Signal N determines if the rider is new. If both conditions 1 and 2 are satisfied, and either condition 3 or condition 4 are satisfied, the request will be connected.

   i. Deduce the logic function using the Boolean signals F, R, G, and N to implement the Juber's algorithm.

   ii. Design a circuit using ideal diode logic gates to implement this function derived in (i).



Ans: (a) $f = w.x + y.z$

Acknowledgement and References

Some of the problems in this set are taken or adapted from the following sources:

1. Sedra, A. S., & Smith, K. C., Microelectronic Circuits, Oxford University Press
2. Neamen, D. A., Microelectronics: Circuit Analysis and Design, McGraw-Hill